

Bias Mitigation in Mental Health Sentiment Analysis Using BERT with Fairness Techniques

Anup Katuwal

Nepal College of Information Technology

Faculty of Management, Pokhara University, Nepal

anup.202854@ncit.edu.np

Abstract— Automated mental health screening using sentiment analysis can help close treatment gaps. However, models built on large pre-training datasets often carry systemic demographic biases. This study looks at reducing bias across three architectures: a BiLSTM network with GloVe embeddings (M1), a standard fine-tuned BERT model (M2), and a fairness-focused version called FairBERT (M3). FairBERT uses adversarial debiasing, counterfactual data augmentation, and focal loss. We ran experiments on RMH-Bias-10K, a specially created dataset of 10,000 Reddit mental health posts. These posts were labeled with binary sentiment tags and four demographic factors: inferred gender, crisis severity, dialect variation, and subreddit community. The agreement between annotators on a 500-post validation set reached a Cohen’s kappa of 0.88. Across five random seeds, M3 achieved 84.0% accuracy and a macro-F1 score of 0.688. It outperformed M1 by 10.3 and 11.7 percentage points, and M2 by 4.5 and 2.8 percentage points, respectively. At the same time, M3 reduced the mean demographic parity difference by 25.6% and the mean equalized odds gap by 15.6% compared to M2. However, dialect-based equalized odds decreased by 9.5%, and there was a persistent false-positive difference between female-inferred and male-inferred posts (0.26 versus 0.14). This shows that aggregate parity metrics can hide imbalances in error rates at the subgroup level..

Keywords— sentiment analysis, BERT, BiLSTM, adversarial debiasing, counterfactual data augmentation, algorithmic fairness, mental health NLP

I. INTRODUCTION

Mental health disorders affect nearly one billion people worldwide, yet treatment infrastructure remains grossly inadequate in most regions [1]. Low- and middle-income countries face treatment gaps exceeding 75 percent [2]. Nepal exemplifies the challenge with particular clarity: according to the 2020 National Mental Health Survey, 13 percent of Nepali adults experience a mental health condition, but only 22.7 percent receive any form of assistance and a mere 3.7 percent access hospital-level services [3]. Structural barriers compound the problem. The country has only 0.06 psychiatrists per 100,000 people, difficult terrain keeps remote communities without services apart, and pervasive stigma of mental illness makes people reluctant to seek help. These figures are not unique to Nepal; they represent a pattern observed across South Asia and many other low-resource settings where the gap

between mental health need and available professional support has remained stubbornly wide.

Given these constraints, researchers have begun exploring what can be done with the data people are already generating. Reddit hosts some of the largest online mental health communities, where users share experiences of distress, coping, and recovery with a candor rarely seen in clinical settings [4]. Subreddits such as r/depression, r/SuicideWatch, and r/Anxiety produce substantial volumes of text daily, and the language within these posts contains genuine markers of emotional states. The research question that emerged from this observation was whether automated tools could read this text at scale and identify individuals who might require support [5], [6]. The volume and first-person nature of these disclosures make social media a uniquely rich data source for population-level mental health monitoring.

Early attempts at automated sentiment analysis used fixed lexicons that assign predetermined polarity scores to words. SentiWordNet [7] and VADER [8] were among the most widely adopted tools. These approaches work reasonably well on clean, edited text, but they struggle with the language of mental health communities. The word empty, for instance, is lexically neutral in most dictionaries yet frequently signals severe depression in a clinical post. Sarcasm, dark humor, and coded expressions of distress—all common in these communities—are routinely misclassified or ignored entirely [9]. Adapting these tools to a new domain requires manual reconstruction of the dictionary, which is neither scalable nor reliable. Statistical machine learning methods based on bag-of-words representations improved upon rule-based systems by learning from labeled data, but they discard word order and therefore miss negation, intensification, and tonal shifts within a single sentence.

Recurrent neural networks addressed the sequence problem directly. Long Short-Term Memory (LSTM) networks [10] maintain internal states across a sequence, and Bidirectional LSTMs process input in both forward and backward passes so that every position has full context. Combining BiLSTMs with pre-trained GloVe word embeddings [11], which encode semantic similarity learned from billions of tokens, showed meaningful improvements over bag-of-words baselines on mental health classification tasks [5], [12]. Attention mechanisms that weight positions by their

importance to the prediction further enhanced performance. BiLSTM with attention represented the state of the art in mental health text classification between approximately 2015 and 2018 and serves as the M1 baseline in this paper.

Transformer-based models fundamentally altered the landscape. BERT [13], pre-trained on approximately 3.3 billion words using masked language modeling and then fine-tuned on task-specific data, outperformed all prior approaches on NLP benchmarks. Mental health researchers rapidly adopted it: depression detection [14], suicide risk screening [15], and emotional state classification [16] all showed gains when BERT replaced earlier architectures. The trend was consistent enough across studies that BERT became the default starting point for this type of work. Subsequent variants including RoBERTa and XLNet further improved upon the original training configuration, though BERT-base remains the most studied and reported model and is therefore selected as the transformer baseline in this study.

The problem is that BERT does not only learn language. It acquires the distributional patterns of its training data, including their demographic skews. Multiple studies have documented that BERT-family models produce systematically different outputs when processing text associated with different demographic groups [17], [18], [19], [20]. In mental health screening, such disparities translate directly into missed cases: when a model underestimates distress in posts written by certain demographics or in non-standard English, those users are effectively invisible to the system. There is no backup. In settings like Nepal, where clinical professionals are so scarce, an automated system that is biased cannot rely on downstream human review to catch its mistakes.

Although algorithmic fairness research in NLP has grown rapidly [21], [22], the majority of it has concentrated on high-resource Western settings. Techniques such as adversarial debiasing [23] and counterfactual data augmentation (CDA) [24] have proven effective in general NLP tasks, but none had been rigorously evaluated on mental health text before this study. Mental health language is emotionally saturated, context-dependent, and crisis-variable in ways that standard NLP benchmarks do not reflect. Whether bias mitigation methods transfer effectively to this domain was a genuinely open question.

This paper makes three contributions. First, it introduces RMH-Bias-10K, a corpus of 10,000 Reddit mental health posts annotated with binary sentiment labels and demographic metadata across four protected attributes: inferred gender, crisis severity, dialect variation, and subreddit affiliation. Second, it presents a controlled comparison of three architectures: BiLSTM with GloVe (M1), vanilla BERT (M2), and FairBERT (M3) combining adversarial debiasing, CDA, and focal loss. Third, it provides empirical evidence that fairness-aware training can reduce demographic disparities without sacrificing classification performance, while carefully documenting the limits, trade-offs, and subgroup-level residual inequities that persist despite overall improvement.

II. RELATED WORK

A. Sentiment Analysis in Mental Health NLP

Computational monitoring of mental health through social media text has been an active research area since the early 2010s. The foundational premise is that language patterns in user posts carry quantifiable indicators of mental state, making passive large-scale screening at least theoretically achievable. Coppersmith et al. [12] demonstrated on Twitter data that linguistic features alone suffice to distinguish self-reported depressed users from matched controls at approximately 70 percent accuracy. De Choudhury et al. [5] conducted a longitudinal study and identified measurable changes in language, posting frequency, and social interaction in the months preceding a first depression diagnosis. The trends were consistent: increased first-person singular pronoun use, more negative-emotion words, shorter sentences, and reduced discussion of social topics.

Reddit subsequently became the preferred platform for deeper analysis. Compared to the character-limited messages on Twitter, long-form posts provide richer linguistic data, and mental health subreddits organized around specific conditions offer researchers interpretable labels based on community membership. The CLPsych shared task series [15] established standard benchmarks, including a 2019 task that involved categorizing users into four levels of suicide risk. Winning systems employed BERT encoders with behavioral metadata, achieving macro-F1 scores between 0.50 and 0.65, which indicate both genuine model capability and the genuine difficulty of fine-grained risk classification from text alone. Demszky et al. [16] released GoEmotions, a 58,000-comment dataset with 27 emotion categories. Although not mental health specific, it has been used to pre-train emotion classifiers that are subsequently adapted to clinical tasks through transfer learning.

Persistent challenges pervade this literature. Class imbalance is perhaps the most steady: even in crisis-oriented communities, severely distressed posts constitute a minority of total content, meaning a model that predicts only the majority class can achieve misleadingly high accuracy while failing to identify the at-risk individuals it was designed to detect. Cultural and linguistic specificity is the second major problem. Pre-trained models reflect the language of their training data, which is heavily skewed toward formal Standard American or British English. Community-specific vocabulary, non-standard dialect features, and culturally inflected expressions of distress are not handled well without explicit domain adaptation [22]. The ethical dimensions of this work have also received increasing attention. Chancellor et al. [9] catalogued the risks: consent problems arising from analyzing posts without explicit authorization, re-identification risks from supposedly anonymous text, and the probability of harm if predictions are acted upon without proper human oversight.

B. Bias in Pre-Trained Language Models

The recognition that large pre-trained language models carry demographic biases has evolved from a single observation to one of the most critical research areas in NLP. The earliest systematic evidence came from static word embeddings. Bolukbasi et al. [18] demonstrated that word2vec vectors encode gender stereotypes geometrically: the difference between man and woman vectors yields a consistent gender direction in the embedding space, and occupation words cluster along that direction in a pattern that closely mirrors cultural stereotypes. The same patterns were observed in GloVe embeddings [25], and subsequent work expanded the analysis to race, nationality, religion, and age.

When BERT arrived, it was not obvious whether contextual training would diminish these effects. Counterfactual testing soon showed it did not. Bhardwaj et al. [17] created sentence pairs identical except for a demographic marker—substituting a male name with a female name or swapping a gendered pronoun—and measured the resulting shift in model predictions. Systematic shifts indicate that the model relies on demographic indicators, rather than semantic content, to make its predictions. Zhao et al. [20] contributed an additional finding: models do not merely reflect biases in training data but amplify them, because biased associations reduce training loss and therefore receive stronger gradient reinforcement during optimization. This phenomenon, termed bias amplification, is directly relevant to interpreting the demographic parity results in this study.

Dialect bias deserves particular mention due to its direct connection to the fairness metrics examined here. Blodgett et al. [22] demonstrated that NLP sentiment classifiers perform substantially worse on African American Vernacular English than on Standard American English. The disparity stems from two related factors: non-standard varieties are underrepresented in pre-training corpora, and standard benchmarks lack subgroup evaluation so the disparity goes unmeasured. In mental health applications, this means a model may miss distress signals written in non-standard English—not because the distress is less real, but because the language pattern is less familiar to the model. Leavy [26] situates these technical findings in a broader argument: bias in NLP reflects not just data problems but systematic inequities in who builds systems and whose needs are treated as normative.

C. Debiasing and Mitigation Strategies

The fairness literature classifies mitigation methods by their point of intervention in the machine learning pipeline: pre-processing, in-processing, or post-processing [27]. Pre-processing approaches modify the training data before it reaches any model. Counterfactual Data Augmentation (CDA) [18], [24] generates training pairs that differ only in a demographic characteristic, such as swapping male pronouns for female pronouns in otherwise identical sentences. When both versions are present, the model cannot learn to use that marker as a predictor. The limitation of CDA is that it addresses only those markers explicitly specified; less obvious demographic cues such as topic associations, writing style, and vocabulary preferences remain intact.

In-processing approaches modify the training objective itself. Adversarial debiasing [23] trains a primary classifier alongside an adversary that attempts to extract demographic attributes from the same learned representations. A gradient reversal layer inverts the direction of adversary gradients during backpropagation, creating a minimax game where the encoder learns to be simultaneously informative for the primary task and uninformative for demographic prediction. Relative to CDA, adversarial debiasing does not require specifying which markers to target—the adversary can discover any leaking signal—but the minimax objective is sensitive to learning rates and adversary capacity, and convergence is not guaranteed. Post-processing approaches adjust decision thresholds at inference time without modifying the model. Hardt et al. [28] showed that equalized odds can be achieved through group-specific threshold optimization, but this requires group membership information at prediction time and does nothing to address biased internal representations.

A foundational theoretical result by Barocas et al. [27] provides essential context for interpreting the experimental findings of this study. They demonstrated that when base rates differ between groups, three common fairness criteria—demographic parity, equalized odds, and calibration—cannot in general be satisfied simultaneously. This is a mathematical constraint, not a shortcoming of any particular technique. Practitioners must therefore choose which criterion matters most for their deployment context and accept that optimizing one may degrade another. This paper evaluates all three criteria and reports the observed trade-offs rather than presenting only the metrics that appear favorable.

III. DATASET: RMH-BIAS-10K

A. Motivation and Design

The central obstacle to studying fairness in mental health NLP is the absence of demographically annotated data. Existing corpora for mental health sentiment and risk classification, including those used in the CLPsych shared tasks [15] and the UMD Reddit Suicidality Dataset [29], contain no information about the demographics of the poster. Without demographic metadata, it is impossible to measure whether a model performs differently across groups, assess whether specific populations are more vulnerable to certain types of bias, or determine whether a given mitigation strategy produces genuine fairness gains. RMH-Bias-10K was constructed to address this gap, combining sentiment annotations with multi-dimensional demographic metadata while meeting academic reliability standards through formal inter-annotator agreement measurement and independent expert validation.

B. Data Collection and Cleaning

Posts were collected from ten mental health subreddits: r/depression, r/Anxiety, r/mentalhealth, r/SuicideWatch, r/BipolarReddit, r/PTSD, r/OCD, r/Lonely, r/selfharm, and r/offmychest, spanning the period from January 2020 to December 2024. These communities were selected based on three criteria: each community focuses on first-person lived

experience of mental health conditions rather than academic discussion or clinical information sharing; each community is sufficiently active to provide adequate data volume; and the set as a whole covers a range of conditions, crisis severity levels, and community cultures to support meaningful subgroup analysis.

Data collection proceeded in two phases due to changes in Reddit’s data access policies. Historical posts from 2020 to 2023 were retrieved via the Pushshift Reddit archive [30], which provided a comprehensive index of submissions with full text. Posts from May 2023 to December 2024 were collected through the official Reddit API via the Python Reddit API Wrapper (PRAW). Only self-authored submissions of at least 50 characters in English (detected via langdetect with confidence threshold 0.95) were retained. Posts that were deleted, removed by moderators, or authored by bot accounts were excluded. A systematic cleaning pipeline removed URLs, HTML tags, username mentions, and Markdown formatting characters while preserving linguistically meaningful content and emotional indicators. Posts dominated by quoted text (over 30 percent) or structured clinical information were filtered out. After cleaning and exclusions, the annotation-eligible pool contained approximately 12,000 posts.

C. Two-Stage Annotation Protocol

Sentiment annotation followed a two-stage design balancing annotation quality with scale. Stage 1 involved manual labeling of 5,000 posts sampled to ensure equal representation across the ten subreddits. A binary schema was used: Negative (distress, hopelessness, suicidal ideation, self-harm, crisis support requests, or significant mental health decline) and Positive (hope, recovery progress, gratitude, encouragement, active coping, or meaningful improvement). True neutral posts were excluded to ensure clean class boundaries. For mixed-emotion posts, the overall emotional trajectory of the entire post was used rather than labeling individual sentences. Dark humor or irony expressing underlying distress was coded by the actual emotional state rather than surface tone.

An independent annotator, a Master’s student in Psychology serving as a practicing student counselor, validated a stratified random sample of 500 posts from Stage 1. The annotator received only the written schema and operational definitions, with no access to original labels. The two annotators agreed on 487 of 500 posts, yielding 97.4 percent observed agreement. After chance correction using Cohen’s kappa with the observed class distribution (88.1 percent Negative, 11.9 percent Positive), the kappa coefficient reached 0.88, falling within the almost-perfect agreement range [31]. The 13 disagreements involved dark-humor posts where ironic surface language concealed real distress and mixed-coping posts where acute distress coexisted with active recovery. These were resolved through a structured reconciliation session with documented reasoning.

Stage 2 extended coverage to the remaining posts using a calibrated automated pipeline combining domain-specific

keywords (identified during manual annotation) and VADER compound scores [8]. Thresholds and keyword weights were calibrated on a held-out subset of Stage 1 data to maximize consistency with gold-standard labels. Posts generating low-confidence or conflicting outputs were filtered. The final RMH-Bias-10K corpus comprises 10,000 posts with an 88.1 percent negative to 11.9 percent positive class ratio, split into 70 percent training (7,000), 15 percent validation (1,500), and 15 percent test (1,500) partitions using stratified random sampling.

D. Demographic Metadata Layer

Four protected attributes were inferred at the post level using textual cues, since Reddit users do not self-report demographics in a structured format. Inferred gender was derived from pronoun patterns, gendered relational terms, and username analysis, yielding 18.5 percent male-inferred, 16.8 percent female-inferred, and 64.7 percent unknown. The binary scheme does not represent non-binary or transgender identities, which is an acknowledged limitation. Dialect variation was classified as Standard or Non-Standard English (74.2 percent versus 25.8 percent) based on systematic grammatical features associated with recognized dialect varieties and consistent non-standard spelling patterns. Crisis severity was determined through VADER scoring and crisis keyword matching (32.4 percent High, 67.6 percent Low). All labels are proxy inferences rather than validated ground-truth demographics. The fairness evaluation is interpreted as measuring systematic model behavioral differences associated with textual characteristics that correlate with demographic groups, not as definitive statements about how specific individuals experience bias.

TABLE I: DATASET STATISTICS BY SPLIT

Split	Total	Neg.	Pos.	Neg.%
Train	7,000	6,167	833	88.1%
Val.	1,500	1,322	178	88.1%
Test	1,500	1,321	179	88.1%
Total	10,000	8,810	1,190	88.1%

E. Comparison with Existing Datasets

The closest existing resources, the CLPsych 2019 dataset (approximately 500 users), the UMD Reddit Suicidality Dataset (approximately 10,000 users), and GoEmotions (58,000 comments), are all Reddit-based but none includes demographic annotations. The fairness assessment conducted in this study cannot be replicated on any existing corpus. RMH-Bias-10K bridges this gap by combining a domain-focused mental health corpus with multi-dimensional demographic metadata. The trade-offs must be stated explicitly: the binary sentiment schema is less clinically granular than four-level risk classification, the demographic labels are inferred rather than self-reported, and the scope is limited to English-language Reddit content.

IV. METHODOLOGY

A. Experimental Design

The experimental framework employs a controlled within-subject comparison with model architecture as the primary independent variable and fairness intervention as the secondary one. Three models are evaluated: M1 (BiLSTM with GloVe embeddings and attention), M2 (vanilla BERT fine-tuned for binary sentiment classification), and M3 (FairBERT, extending M2 with adversarial debiasing, counterfactual data augmentation, and focal loss). Each model is trained five times with different random seeds (42, 123, 456, 789, 1024), producing fifteen experimental runs total. This multi-seed design enables computation of means and standard deviations and supports paired t-tests for statistical significance. All models share identical data splits, preprocessing pipelines, evaluation protocols, and hardware configurations; only the architecture and fairness interventions differ, ensuring that observed performance differences can be attributed to these factors.

Dependent variables fall into two categories. Performance variables include accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, and AUC-ROC, all computed on the held-out test set. Fairness variables include demographic parity difference (DPD), equalized odds gap (EOG), and accuracy parity gap (APG), computed separately for each of the four protected attributes. This dual assessment in both categories is the distinguishing methodological feature of this study relative to most existing mental health NLP work, which reports only aggregate performance.

B. Data Preparation and Class Imbalance

Text preprocessing converts all posts to lowercase, removes URLs and HTML tags, compresses repeated punctuation to single instances, and normalizes whitespace while retaining grammatical structures and emotive intensifiers. For M3 exclusively, counterfactual data augmentation expands the training set by generating gender-swapped variants using word boundary-constrained substitution (regex `\b` anchors) of ten gendered term pairs including he/she, him/her, boyfriend/girlfriend, husband/wife, father/mother, brother/sister, and son/daughter. Posts containing at least one gender marker produce one counterfactual variant, resulting in an augmented training set approximately 1.8 times the original size. Augmentation is applied only to training data; validation and test sets remain unmodified so that fairness metrics reflect real distributional properties rather than artificially balanced data.

The 88.1 percent to 11.9 percent class ratio presents a genuine training challenge: an uncorrected classifier achieves 88.1 percent accuracy by predicting only the majority class without learning anything useful. Two countermeasures are shared across all models: random undersampling of the negative class to approximately 4:1 and class-weighted loss with positive-class weight of approximately 7.4 (the inverse of its frequency). M3 additionally employs focal loss [32] on the sentiment classification objective. The focal loss modulation factor scales down contributions from confidently classified

examples and redirects training capacity toward harder, ambiguous cases. With the focusing parameter set to 2.0, the loss from a confidently classified example is reduced by up to a factor of four relative to a hard case. In an imbalanced setting where the model can plateau early by learning the majority-class boundary, this focusing pressure prevents training from stalling before the minority class is adequately learned.

C. Model Architectures

1) M1: BiLSTM with Attention Mechanism:

M1 maps words to 300-dimensional frozen GloVe vectors trained on 840 billion tokens, processes sequences through two stacked bidirectional LSTM layers with hidden size 128 per direction, applies a learned attention mechanism to compute position-weighted context vectors, and projects through dropout ($p = 0.3$) to binary sentiment predictions. The attention weights are computed as the softmax of a tanh-transformed linear projection of the hidden states, and the context vector is the weighted sum of those states. Training uses Adam with learning rate 0.001, batch size 32, maximum sequence length 256 tokens, gradient clipping at norm 1.0, and early stopping with patience of 3 epochs on validation macro-F1.

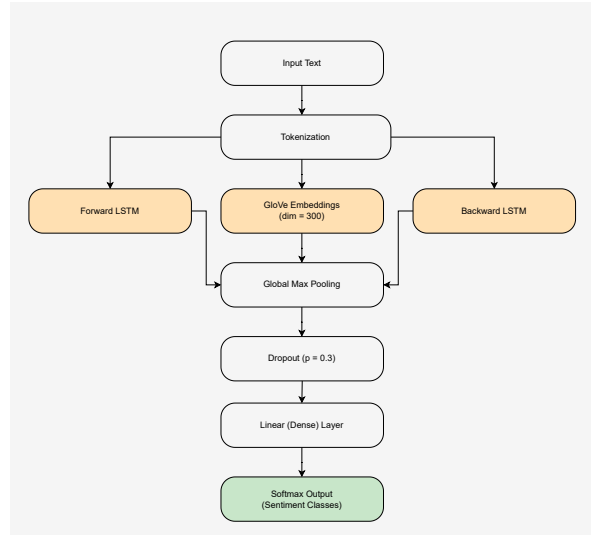


Fig. 1 BiLSTM with GloVe embedding architecture

2) M2: Vanilla BERT for Sentiment Classification:

M2 fine-tunes bert-base-uncased (110 million parameters, 12 transformer layers) for binary sentiment classification without any fairness-specific modifications. Input text is tokenized into WordPiece subword units with a maximum sequence length of 128 tokens. The 12 transformer layers of multi-head self-attention and feed-forward networks produce contextual representations at each position, with residual connections and layer normalization at each layer. For classification, the final hidden state of the [CLS] token serves as the sequence representation and is projected through a linear layer to binary sentiment logits. Training uses AdamW with learning rate 2×10^{-5} to the negative fifth, weight decay 0.01, 10 percent warmup, batch size 16, and weighted cross-entropy loss with early stopping on validation macro-F1.

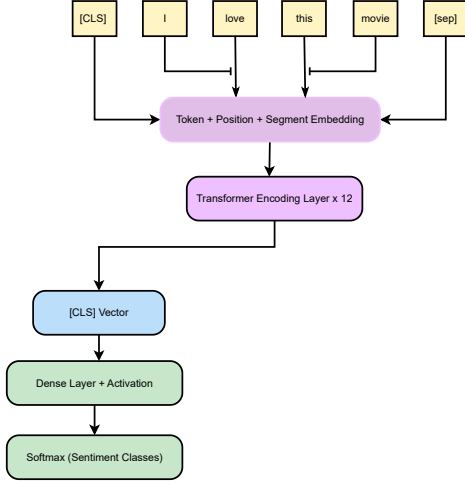


Fig. 2 BERT architecture with sentiment classification head attached.

3) M3: FairBERT with Adversarial Debiasing:

FairBERT extends M2 with two fairness interventions: counterfactual data augmentation at the data level and adversarial debiasing at the model level. The adversarial component trains a primary sentiment classifier alongside a discriminator that attempts to predict inferred gender from the shared BERT representations. A gradient reversal layer connects the encoder to the adversary, inverting gradient directions during backpropagation. The total training loss combines the sentiment classification objective (focal loss) with the adversarial objective:

$$L_{total} = L_{sentiment} - \lambda \times L_{adversary} \quad (1)$$

where λ controls the debiasing strength and is annealed linearly from 0 to 0.5 over the first training epoch. The negative sign creates the minimax game: the encoder minimizes sentiment error while maximizing the adversary's error, thereby learning representations that encode sentiment but obscure gender. The adversary uses a 2-layer feedforward network with ReLU activations, trained with Adam at learning rate 1×10^{-4} to the negative fourth. Gender was selected as the adversarial target because it is the most extensively studied dimension in the debiasing literature and provides the strongest available training signal given the binary annotation scheme. Fairness improvements on other attributes—dialect, crisis severity, and subreddit community—are evaluated as downstream effects of the combined gender-targeted debiasing and counterfactual augmentation.

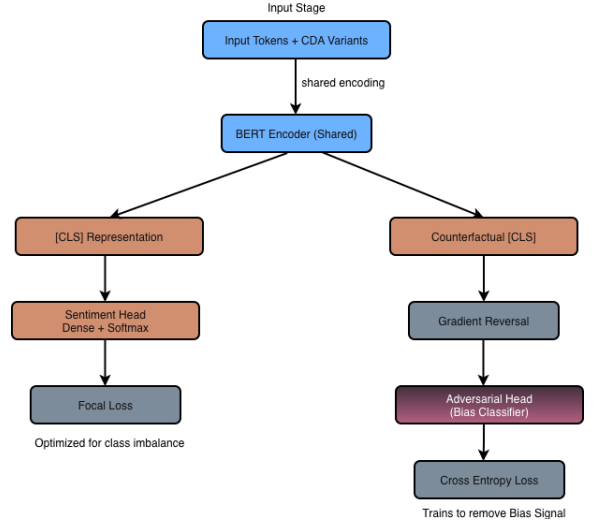


Fig. 3 FairBERT architecture with Adversarial head, CDA, and Focal loss

D. Training Procedure

All three models follow an identical training procedure to ensure comparability. At the beginning of each run, the random seed controls PyTorch, NumPy, and Python random number generators. Pre-trained weights are loaded from the HuggingFace Transformers library checkpoint bert-base-uncased for M2 and M3, and from the pre-trained 840-billion-token GloVe file for M1. Each model trains for up to 20 epochs with gradient clipping at maximum norm 1.0. After each epoch, the model is evaluated on the validation set and the macro-F1 score is recorded. If validation macro-F1 fails to improve for three consecutive epochs, training halts and the best checkpoint is restored for final test evaluation. Each configuration runs five times with seeds 42, 123, 456, 789, and 1024. Hyperparameters are fixed across seeds; only initialization differs. All experiments were executed on Google Colab Pro+ using NVIDIA T4 and A100 GPU instances. M1 completed its five-seed cycle in approximately 45 minutes, M2 in 90 minutes, and M3 in roughly 3 hours, reflecting the additional computational overhead of adversarial training.

E. Evaluation Metrics

Classification performance is measured by accuracy, macro-averaged precision, recall, F1-score, and AUC-ROC. Macro-F1 serves as the primary metric because it weights both classes equally regardless of frequency, making it appropriate for imbalanced datasets. Fairness is assessed using three complementary metrics computed for each protected attribute. Demographic Parity Difference (DPD) measures absolute differences in positive prediction rates across groups; values below 0.1 are conventionally considered to indicate operational fairness. Equalized Odds Gap (EOG) measures maximum differences in true positive rates (TPR) and false positive rates (FPR) across groups, requiring equal error rates regardless of group membership. Accuracy Parity Gap (APG) measures differential accuracy amplification, comparing the gap between predicted and actual positive rates across groups. Lower values indicate greater fairness for all metrics, though perfect fairness

across all criteria simultaneously is generally impossible when base rates differ between groups [33].

V. RESULTS AND ANALYSIS

A. Overall Performance

TABLE II: OVERALL PERFORMANCE (MEAN $\pm$ STD DEV, 5 SEEDS)

Metric	M1	M2	M3	Best
Acc.	.737 $\pm$.054	.795 $\pm$.012	.840 $\pm$.024	M3
Prec.	.575 $\pm$.013	.640 $\pm$.006	.669 $\pm$.018	M3
Recall	.631 $\pm$.022	.750 $\pm$.008	.736 $\pm$.020	M2
F1	.571 $\pm$.021	.660 $\pm$.008	.688 $\pm$.015	M3
AUC	.708 $\pm$.014	.839 $\pm$.006	.843 $\pm$.005	M3

M3 achieves a mean accuracy of 0.840 (SD = 0.024), representing improvements of 10.3 percentage points over M1 and 4.5 percentage points over M2. The macro-F1 of 0.688 (SD = 0.015) reflects well-balanced precision-recall performance, with gains of 11.7 percentage points over M1 and 2.8 percentage points over M2. The AUC-ROC of 0.843 (SD = 0.005) indicates strong discriminative capacity, representing a 13.5 percentage point improvement over M1 and a marginal 0.4 percentage point gain over M2. M2 attains the highest macro recall (0.750 versus 0.736 for M3), reflecting a more liberal classification threshold that raises sensitivity at the cost of precision. M3’s superior macro-F1 indicates that its precision-recall balance is better calibrated for the evaluation criterion used in this study.

B. Statistical Significance Testing

Paired t-tests across five seeds confirm that M3’s advantages are not attributable to random seed variation. Against M1, all four primary metrics achieve significance at the $\alpha = 0.05$ threshold with large effect sizes: accuracy ($\Delta = +0.103$, $t = 3.31$, $p = 0.030$, Cohen’s $d_z = 1.48$), macro-F1 ($\Delta = +0.117$, $t = 5.28$, $p = 0.006$, $d_z = 2.36$), positive-class F1 ($\Delta = +0.164$, $t = 4.76$, $p = 0.009$, $d_z = 2.13$), and AUC-ROC ($\Delta = +0.135$, $t = 18.10$, $p < 0.001$, $d_z = 8.09$). The AUC-ROC result is particularly striking—a d_z of 8.09 indicates that M3’s discriminative advantage over M1 was essentially identical across all five seeds. Against M2, accuracy ($t = 3.38$, $p = 0.028$, $d_z = 1.51$) and macro-F1 ($t = 3.60$, $p = 0.023$, $d_z = 1.61$) reach significance with large effect sizes. The AUC-ROC difference does not reach significance ($p = 0.254$), which is expected given that both BERT-based models already achieve strong discrimination.

C. Per-Class and Confusion Matrix Analysis

TABLE III: PER-CLASS PERFORMANCE (MEAN $\pm$ STD DEV)

Class	Metric	M1	M2	M3
Neg.	Prec.	.904 $\pm$.023	.950 $\pm$.005	.953 $\pm$.007
	Recall	.779 $\pm$.053	.825 $\pm$.012	.853 $\pm$.018
	F1	.835 $\pm$.033	.883 $\pm$.008	.900 $\pm$.011
Pos.	Prec.	.247 $\pm$.033	.330 $\pm$.015	.386 $\pm$.035
	Recall	.483 $\pm$.052	.675 $\pm$.021	.618 $\pm$.046

	F1	.307 $\pm$.029	.437 $\pm$.018	.471 $\pm$.021
--	----	-----------------	-----------------	-----------------

All models achieve strong negative-class performance, with M3 reaching F1 = 0.900 (SD = 0.011). The positive class poses significantly stricter requirements due to its 11.9 percent prevalence and the inherent linguistic ambiguity between positive sentiment and the mere absence of acute distress in a mental health context. M3 achieves a positive-class F1 of 0.471 (SD = 0.021), representing a 53.4 percent relative improvement over M1 (0.307) and a 7.8 percent relative improvement over M2 (0.437). The positive-class recall of M2 (0.675) exceeds M3 (0.618), but M2’s positive-class precision is substantially lower (0.330 versus 0.386), indicating that M2 generates approximately 35 more false positives per evaluation run. At representative seed 42, M3 produces 1,141 true negatives (86.3 percent specificity), 116 true positives (65.2 percent sensitivity), 181 false positives, and 62 false negatives. Whether higher recall or higher precision is preferable depends on the asymmetric costs of false negatives and false positives in the target deployment scenario; under the macro-F1 criterion used for model selection, M3 achieves the best-calibrated operating point.

D. Fairness Evaluation: Demographic Parity

TABLE IV: DEMOGRAPHIC PARITY DIFFERENCE (MEAN $\pm$ STD DEV)

Attribute	M1	M2	M3	Improv.
Gender	.191 $\pm$.079	.305 $\pm$.045	.198 $\pm$.068	-35.1%
Crisis	.188 $\pm$.038	.280 $\pm$.032	.211 $\pm$.055	-24.6%
Dialect	.042 $\pm$.014	.053 $\pm$.020	.037 $\pm$.025	-30.5%
Subreddit	.362 $\pm$.052	.442 $\pm$.042	.360 $\pm$.071	-18.5%
Mean	.196$\pm$.046	.270$\pm$.034	.201$\pm$.054	-25.6%

M3 reduces mean DPD by 25.6 percent relative to M2. The strongest improvement occurs on inferred gender (35.1 percent reduction, from 0.305 to 0.198), reflecting the adversarial component’s direct targeting of gender signals. Crisis severity and subreddit DPD decrease by 24.6 and 18.5 percent, respectively, while dialect DPD remains consistently below the 0.1 fairness threshold across all models. Notably, M2 exhibits higher mean DPD than M1 (0.270 versus 0.196), providing empirical support for the hypothesis that large-scale pre-training on uncured internet corpora can amplify rather than reduce demographic biases in domain-specific applications. The subreddit attribute shows the largest absolute DPD values (M2 = 0.442), likely reflecting genuine differences in how mental health communities express sentiment, which no model fully normalizes.

E. Fairness Evaluation: Equalized Odds

TABLE V: EQUALIZED ODDS GAP (MEAN $\pm$ STD DEV)

Attribute	M1	M2	M3	Improv.
Gender	.241 $\pm$.118	.334 $\pm$.081	.244 $\pm$.092	-26.9%
Crisis	.190 $\pm$.021	.225 $\pm$.032	.159 $\pm$.044	-29.3%
Dialect	.183 $\pm$.061	.283 $\pm$.074	.310 $\pm$.055	+9.5%
Subreddit	.579 $\pm$.116	.466 $\pm$.049	.393 $\pm$.087	-15.6%

Mean	.298±.079	.327±.060	.276±.069	-15.6%
------	-----------	-----------	-----------	--------

M3 achieves a mean EOG of 0.276 (SD = 0.069), a 15.6 percent improvement over M2. The strongest gains appear for crisis severity (29.3 percent reduction) and gender (26.9 percent reduction). However, the dialect attribute presents a striking exception: M3’s dialect EOG increases by 9.5 percent to 0.310. This result directly illustrates the impossibility theorems: the interventions that successfully narrow the demographic parity gap for dialect simultaneously widen the error-rate gap between dialect groups. Satisfying one fairness criterion does not guarantee fulfillment of another.

A decomposition into TPR and FPR reveals asymmetrical patterns across gender subgroups. At seed 42, M3 records a female TPR of 0.70 and FPR of 0.26, whereas the male group shows TPR = 0.66 and FPR = 0.14. In a practical screening context, this asymmetry creates a differential false-alarm burden: the model identifies positive cases in female-authored posts more readily but simultaneously generates disproportionately more false positives for that group. The unknown gender group’s consistently low TPR (0.40 to 0.54) suggests that posts lacking gender-linked lexical cues are systematically under-detected—an important consideration for any deployment where the majority of users may not provide gendered language markers.

F. Accuracy Parity and Trade-off Analysis

TABLE VI: ACCURACY PARITY GAP (MEAN ± STD DEV)

Attribute	M1	M2	M3	Reduc.
Gender	.158±.056	.187±.027	.142±.031	-24.1%
Crisis	.103±.030	.165±.015	.160±.021	-3.4%
Dialect	.062±.023	.034±.007	.082±.038	+141%
Subreddit	.328±.040	.322±.037	.275±.041	-14.7%
Mean	.163±.037	.177±.022	.165±.033	-6.8%

M3 produces a mean APG of 0.165 (SD = 0.033), a modest 6.8 percent reduction from M2. Gender APG improves from 0.187 to 0.142 (24.1 percent reduction) and subreddit APG from 0.322 to 0.275 (14.7 percent reduction), but dialect APG increases from 0.034 to 0.082, consistent with the dialect EOG regression. The M2-to-M3 comparison is the most informative for assessing trade-offs because both models share the same BERT architecture and differ only in fairness interventions. M3 outperforms M2 simultaneously on accuracy (0.840 versus 0.795), macro-F1 (0.688 versus 0.660), mean DPD (0.201 versus 0.270), and mean EOG (0.276 versus 0.327)—achieving Pareto dominance across both performance and fairness dimensions. This result challenges the common assumption that accuracy and fairness are inherently competing objectives.

G. Cross-Seed Stability and Subgroup Analysis

M2 exhibits the narrowest standard deviations on performance metrics (accuracy SD = 0.012, macro-F1 SD = 0.008), reflecting the stabilizing effect of pre-trained BERT initialization. M3 shows slightly higher variance on

performance (accuracy SD = 0.024) attributable to stochasticity from adversarial training and CDA sampling, but achieves the lowest standard deviations on AUC-ROC (0.005) and mean APG (0.015), indicating that its discriminative capacity and certain fairness behaviors are the most consistent elements of its profile. M1 exhibits substantially higher performance variance than either BERT-based model, confirming the practical instability of BiLSTM approaches at this training set size.

Subgroup analysis under M3 reveals a negative correlation between community-level crisis prevalence and overall accuracy. Low-crisis communities such as r/Anxiety (9.6 percent crisis posts) achieve accuracy above 93 percent, while high-crisis communities such as r/SuicideWatch (49.0 percent) reach only 69.4 percent but with substantially higher TPR (0.821). This pattern suggests an appropriate sensitivity bias in high-crisis settings, though the accompanying FPR (0.347 for r/SuicideWatch) indicates that the model also over-predicts positive sentiment in crisis-heavy contexts. Under the gender attribute, male-inferred posts achieve 82.9 percent accuracy and FPR = 0.146, while female-inferred posts reach only 73.2 percent accuracy with FPR = 0.261. The adversarial component was insufficient to fully equalize gender-subgroup performance, likely constrained by the small number of female-inferred training instances.

VI. DISCUSSION

The central finding of this study, that M3 achieves simultaneous improvements in both predictive performance and demographic fairness, challenges the commonly held assumption that accuracy and equity are inherently opposing forces. The most plausible mechanistic explanation is that the three fairness interventions act synergistically as targeted regularization. Adversarial debiasing penalizes the encoding of spurious demographic associations that would not generalize to unseen data, forcing the model to rely on genuinely sentiment-relevant features. Counterfactual augmentation increases training diversity by exposing the model to systematic variation in gendered language, reducing overfitting to demographic-specific patterns. Focal loss concentrates training capacity on hard, ambiguous examples near the decision boundary—precisely the cases where demographic bias is most likely to manifest as misclassification. Together, these mechanisms appear to have improved both generalization and fairness rather than trading one for the other.

The dialect results provide a critical counterpoint to the overall positive narrative. While M3 successfully narrows the demographic parity gap for dialect (from 0.053 to 0.037), it simultaneously widens the equalized odds gap (from 0.283 to 0.310) and accuracy parity gap (from 0.034 to 0.082). This outcome directly illustrates the impossibility results documented by Barocas et al. [27]: equalizing prediction rates across groups does not guarantee equal error rates, and optimizing one fairness criterion can degrade another. Practitioners deploying fairness-aware models must therefore specify which criterion takes priority in their deployment

context. In a mental health screening application where equal detection rates across dialect groups are prioritized, additional dialect-specific interventions beyond those tested here would be necessary.

The persistent gender disparities merit careful interpretation. The female-inferred subgroup’s higher FPR (0.26 versus 0.14 for males) means that female-authored posts about recovery or coping are more frequently misclassified as negative, potentially leading to unnecessary interventions in a deployed system. This disparity persists despite adversarial debiasing directly targeting gender, suggesting that the constraint is primarily one of data quantity: with only approximately 56 female-inferred test posts, both the training signal for fairness optimization and the statistical power of fairness evaluation are severely limited. Notably, M2 exhibits higher mean DPD than the simpler M1 baseline, providing empirical evidence that scaling model capacity through large-scale pre-training can amplify rather than reduce demographic biases in domain-specific applications—a finding with important implications for the uncritical adoption of ever-larger language models.

The practical significance of these findings extends beyond the immediate experimental context. For Nepal, where only 22.7 percent of adults with mental health conditions receive any treatment [3], an automated screening tool that systematically misses certain demographic groups would not bridge the treatment gap but would redistribute it. The evidence presented here suggests that fairness-aware training can mitigate but not eliminate such risks. The entire experimental pipeline was implemented using open-source libraries in Google Colab, demonstrating that the approach is accessible in resource-constrained environments. However, deployment without continuous subgroup monitoring and human oversight would be premature. The error rates that M3 achieves, though improved over baselines, remain too high for fully automated decision-making in a domain where both false positives and false negatives carry real human costs.

VII. LIMITATIONS

Several limitations constrain the generalizability of these findings and must be stated explicitly. The most severe is the gender evaluation: approximately 56 female-inferred test posts constitute a sample too small for statistically reliable subgroup error-rate estimates. The 64.7 percent unknown-gender rate means the evaluated subgroups may not represent the broader population of mental health posters. All four demographic labels are textually inferred rather than self-reported: gender uses a binary scheme that excludes non-binary and transgender identities; dialect classification collapses complex sociolinguistic variation into a binary Standard/Non-Standard distinction; and crisis severity relies on keyword and VADER-based proxies rather than clinical assessment. These are accepted practices in fairness research when self-reported demographics are unavailable [35], but the resulting metrics are approximations—labeling noise may cause some fairness

measures to understate actual disparities while overstating others.

The dataset is English-only and Reddit-specific. The Reddit user base skews young, Western, and technologically oriented, characteristics that limit transferability to other platforms, other languages, or Nepali-language mental health communities, the very context that initially motivated this research. The four demographic attributes are analyzed independently throughout; intersectional effects where compounding occurs across multiple attributes simultaneously remain unexamined because the intersection cells would have too few instances for reliable evaluation. The model comparison is limited to three architectures; other transformer variants such as RoBERTa and XLNet, alternative fairness techniques such as reweighting and post-hoc calibration, and prompt-based debiasing of large language models are beyond the scope of this work. Finally, no clinical validation was performed: all results stem from offline evaluation, no predictions were reviewed by practitioners, and the question of whether technical fairness improvements translate to equitable real-world service delivery remains entirely open.

VIII. CONCLUSION AND FUTURE WORK

This paper introduced RMH-Bias-10K, the first demographically annotated mental health sentiment dataset, and demonstrated through controlled experimentation that FairBERT (M3), combining adversarial debiasing, counterfactual data augmentation, and focal loss, achieves simultaneous improvements in both predictive accuracy and demographic fairness relative to BiLSTM (M1) and vanilla BERT (M2) baselines. M3 reached 84.0 percent accuracy and 0.688 macro-F1 while reducing mean DPD by 25.6 percent and mean EOG by 15.6 percent compared to M2, without any performance sacrifice. However, the dialect-based equalized odds regression of 9.5 percent, the persistent gender FPR disparity (0.26 versus 0.14), and the limitations of proxy demographic labels demonstrate that aggregate fairness improvements do not guarantee subgroup-level equity—a practical instantiation of the impossibility theorems of algorithmic fairness.

The absence of a performance-fairness trade-off in this study is an empirical finding specific to this dataset, class distribution, and combination of techniques. It should not be generalized to suggest that fairness and performance always align or that fairness interventions are universally cost-free. The gender and dialect results demonstrate that data scarcity and criterion incompatibility can override the benefits of even well-designed mitigation strategies.

Future work should pursue four directions. First, expanding the corpus with more balanced demographic representation and self-reported rather than inferred attributes would strengthen fairness evaluation. Creating annotated datasets in Nepali and other regional languages would directly serve the populations where the treatment gap is most severe. Second, intersectional fairness analysis across combinations of protected characteristics is necessary to detect compounded

disparities invisible to single-attribute evaluation. Third, evaluating additional architectures including RoBERTa, XLNet, and large language models with prompt-based debiasing would clarify whether the dynamics observed here generalize across model families and scales. Fourth, and most critically, prospective clinical validation in collaboration with mental health practitioners is essential to determine whether technical fairness gains translate to equitable outcomes for the people these tools are meant to serve. Without such validation, the findings presented here remain a foundation for further research rather than evidence of practical benefit.

ACKNOWLEDGMENT

The author thanks Prof. Dr. Roshan Chitrakar for supervision throughout this research and the independent annotator for expert validation of the dataset labels. The author is also thankful to Master's Program Co-ordinator Mr. Saroj Shakya for his unconditional help and insight in completing this research. This work was conducted at Nepal College of Information Technology under the Faculty of IT department, Pokhara University.

REFERENCES

- [1] World Health Organization, *Mental Health Atlas 2020*. Geneva, Switzerland: WHO, 2021.
- [2] G. Thornicroft et al., "Undertreatment of people with major depressive disorder in 21 countries," *Brit. J. Psychiatry*, vol. 210, no. 2, pp. 119–124, 2017.
- [3] Nepal Health Research Council, "Report of National Mental Health Survey, Nepal 2020," Kathmandu, Nepal: NHRC, 2023.
- [4] M. De Choudhury and S. De, "Mental health discourse on Reddit: Self-disclosure, social support, and anonymity," in *Proc. ICWSM*, 2014, pp. 71–80.
- [5] M. De Choudhury, M. Gamon, S. Counts, and E. Horvitz, "Predicting depression via social media," in *Proc. ICWSM*, 2013, pp. 128–137.
- [6] A. Di Cara, L. Barbaglia, and P. Luceri, "Monitoring mental health through social media: A scoping review," *J. Biomed. Inform.*, vol. 139, art. no. 104290, 2023.
- [7] A. Esuli and F. Sebastiani, "SentiWordNet: A publicly available lexical resource for opinion mining," in *Proc. LREC*, 2006.
- [8] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proc. ICWSM*, 2014.
- [9] S. Chancellor et al., "A taxonomy of ethical tensions in inferring mental health states from social media," in *Proc. FAT**, 2019, pp. 79–88.
- [10] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [11] J. Pennington, R. Socher, and C. Manning, "GloVe: Global vectors for word representation," in *Proc. EMNLP*, 2014, pp. 1532–1543.
- [12] G. Coppersmith, M. Dredze, and C. Harman, "Quantifying mental health signals in Twitter," in *Proc. CLPsych Workshop, NAACL-HLT*, 2014, pp. 51–60.
- [13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [14] T. Ilias, A. S. M. N. Islam, and M. Hoque, "Calibrating transformer-based models for stress and depression detection from social media," *IEEE Access*, vol. 11, pp. 15832–15845, 2023.
- [15] C. C. MacAvaney et al., "The CLPsych 2019 shared task: Predicting suicide risk from Reddit posts," in *Proc. CLPsych Workshop, NAACL-HLT*, 2019, pp. 24–33.
- [16] D. Demszky et al., "GoEmotions: A dataset of fine-grained emotions," in *Proc. ACL*, 2020, pp. 4040–4054.
- [17] R. Bhardwaj, N. Majumder, and S. Poria, "Investigating gender bias in BERT," *Cogn. Comput.*, vol. 13, no. 4, pp. 1008–1018, 2021.
- [18] T. Bolukbasi et al., "Man is to computer programmer as woman is to homemaker? Debiasing word embeddings," in *Proc. NeurIPS*, 2016, pp. 4349–4357.
- [19] R. Bhardwaj, K. Ghosh, and A. Dey, "Analyzing bias in pre-trained language models for affective tasks," *Front. Artif. Intell.*, vol. 4, p. 789603, 2021.
- [20] J. Zhao et al., "Men also like shopping: Reducing gender bias using corpus-level constraints," in *Proc. EMNLP*, 2017, pp. 2979–2989.
- [21] A. Jorgensen, D. Hovy, and A. Sogaard, "Challenges of studying and processing dialects in social media," in *Proc. Workshop on Noisy User-Generated Text*, 2015, pp. 9–18.
- [22] S. L. Blodgett, L. Green, and B. O'Connor, "Demographic dialectal variation in social media: A case study of African-American English," in *Proc. EMNLP*, 2016, pp. 1119–1130.
- [23] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. ICML*, 2015, pp. 1180–1189.
- [24] D. Kaushik, E. Hovy, and Z. Lipton, "Learning the difference that makes a difference with counterfactually augmented data," in *Proc. ICLR*, 2020.
- [25] N. Garg et al., "Word embeddings quantify 100 years of gender and ethnic stereotypes," *Proc. Nat. Acad. Sci.*, vol. 115, no. 16, pp. E3635–E3644, 2018.
- [26] M. Leavy, "Gender bias in artificial intelligence: The need for diversity and gender theory in machine learning," in *Proc. IEEE/ACM Int. Workshop Gender Equality Softw. Eng.*, 2018, pp. 14–16.
- [27] S. Barocas, M. Hardt, and A. Narayanan, *Fairness and Machine Learning*. Cambridge, MA: MIT Press, 2023.
- [28] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Proc. NeurIPS*, 2016, pp. 3315–3323.
- [29] P. Resnik et al., "UMD Reddit Suicidality Dataset (Version 2)," Univ. Maryland, 2019.
- [30] J. Baumgartner et al., "The Pushshift Reddit Dataset," *arXiv:2001.08435*, 2020.
- [31] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychol. Bull.*, vol. 76, no. 5, pp. 378–382, 1971.
- [32] T.-Y. Lin et al., "Focal loss for dense object detection," in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.
- [33] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent trade-offs in the fair determination of risk scores," in *Proc. ITCS*, 2017, pp. 43:1–43:23.
- [34] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- [35] M. Sap et al., "Social bias frames: Reasoning about social and power implications of language," in *Proc. ACL*, 2020, pp. 5477–5490.
- [36] E. M. Bender and B. Friedman, "Data statements for natural language processing: Toward mitigating system bias and enabling better science," *Trans. Assoc. Comput. Linguist.*, vol. 6, pp. 587–604, 2018.